TECHNISCHE HOCHSCHULE MITTELHESSEN

CAMPUS GIESSEN

MNI
Mathematik, Naturwissenschaften und Informatik

Master Thesis

Configuration-based mobile applications

zur Erlangung des akademischen Grades

Master of Science

eingereicht im Fachbereich Mathematik, Naturwissenschaften und Informatik an der Technische Hochschule Mittelhessen

von

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Erklärung der Selbstständigkeit

Hiermit versichere ich, die vorliegende Arbeit selbstständig verfasst und keine anderen als die angegebenen Quellen und Hilfsmittel benutzt sowie die Zitate deutlich kenntlich gemacht zu haben.

Gießen, den 8. Januar 2019 Samuel Schepp

This thesis describes configuration-based mobile applications and their infrastructure. In addition, a possible way of implementing a configuration system is discussed. Finally, there is an evaluation of the use case of interpreting and generated apps.

Furthermore, the development process and debugging routine will be assessed and compared to a classic development project.

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1 Introduction

configuration based apps.

how? download layout, data definition, actions, metadata.

all apps use the same container, which is always up to date and uses latest technology.

some metadata has to be baked in on compile time.

1.1 Problem definition

why configuration based?

long waiting times on app store approval

example: customer terminal, city marketing stele: lot of work to redeploy large amount of units.

lot of different apps need same boilerplate code. hard to migrate all to newest technology.

1.2 Motivation

use of native mobile features, camera, background gps, beacons, push notification (with payload). caching, fast ui, good battery life.

no need to depend on third party frameworks, direct use of phone vendor's apis.

1.3 Distinction

1.4 Structure

lorem ipsum

2 Background

lorem

2.1 Fabrik19

lorem

2.2 MOS

lorem

3 Concept

Same day delivery

easy configuration

simple datasources

common app tasks simplified

content driven (wordpress, social media)

reactive form elements

4 Implementation

lorem ipsum

4.1 Nodes

4.2 Layouts

4.3 Dataproxy

4.4 Database

4.5 Actions

5 Evaluation

lorem ispum

5.1 Analysis

lorem ispum

5.1.1 Benefits

lorem ispum

5.1.2 Disadvantages

lorem ispum

5.2 Conclusion

lorem ispum

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